Advanced Computer Graphics (ACG)

Theory Session 6 – Volume Rendering: Absorption-Only Model

2026-2027

Pompeu Fabra University

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Calendar

Week no.	Date	Theory	Practice	Seminar	Deliveries
Week 1	28 Sep 02 Oct	**T1**: Introduction	**P1**: Introduction to A1.1	-	
Week 2	05 Oct 09 Oct	**T2**: Basic Ray Tracing		**S1**: Work on A1	
Week 3	12 Oct 16 Oct	**T3**: Ray Casting, Recursive Ray Tracing	**P2**: Introduction to A1.2		
Week 4	19 Oct 23 Oct	**T4**: Rendering Equation, Monte Carlo Integration and Direct Illumination **T5**: Indirect Illumination and Path Tracing		**S2**: Work on A1	**D.A1**
Week 5	26 Oct 30 Oct	**T6**: Volume Rendering: Absorption-Only Model	**P3**: Introduction to A2.1		
Week 6	02 Nov 06 Nov	**T7**: Volume Rendering: Emission, Absorption and Scattering Model **T8:** Neural Radiance Fields and Topics' Presentation		**S3**: Work on A2	
Week 7	09 Nov 13 Nov	**T9**: **Midterm**	**P4**: Introduction to A2.2		
Week 8	16 Nov 20 Nov		**P5**: Work on A2	**S4**: Work on A2	**D.A2**
Week 9	23 Nov 27 Nov	**T10**: **1st Presentation**	**P6**: Work on FP	-	
Week 10	30 Nov 04 Dec	**T11**: **2nd Presentation** **(First Session)** **T12**: **2nd Presentation** **(Second Session)**		**S5**: Work on FP	**D.FP**

Class Outline

- Motivation and Problem Set-up
- Light-Volume Interactions (Micro-level)
  - Absorption
  - Emission
  - In and Out Scattering
- The Radiative Transfer Equation
- Absorption-Only Model and its Numerical Approximation

Introduction & Motivation

Light-Volume Interactions

- Up to now
  - Scenes composed of collections of surfaces
  - Light rays propagate in vacuum (radiance is constant between two surfaces)
- In real world situations these assumptions can be inexact
  - Fog and smoke attenuate light
  - Light scattered in the atmosphere makes the sky look blue

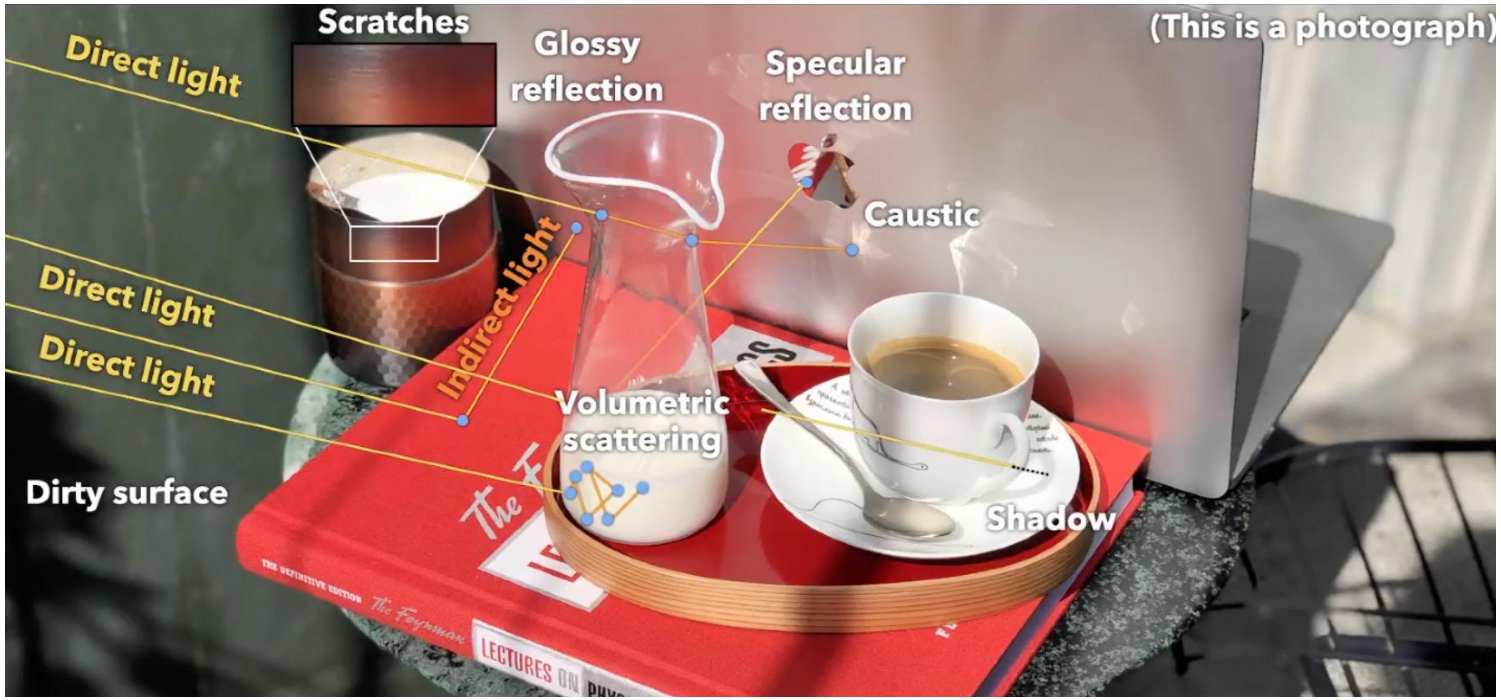


Image from a recent presentation on Physically Based Rendering (by Wenzel Jakob)

Photo-Realistic Image Synthesis

Roadmap

- In this second block of the course, we will:
  - State mathematical formalisms that describe the interaction of light with participating media
  - Define the Radiative Transfer Equation (RTE)
  - Derive the Volume Rendering Equation from the RTE
  - Explore techniques for solving the Volume Rendering Equation



Light-Volume Interactions

Image Credits: Jarosz et al. (2008) “Radiance caching for participating media”



Different Types of Media

- Similarly to surface materials, volumes (more precisely, participating media) exhibit **different appearances and features**
  - Some might be denser
  - Some might emit light
  - Some might reflect more or less light
- In other words, light interacts differently with different types of media (i.e., volumes)
  - Need to characterize the nature of this interaction

Different Types of Media

- **3 main physical processes** occur when light interacts with participating media:
  - **Absorption**: the reduction in radiance due to the conversion of light to another form of energy, such as heat
  - **Emission**: when radiance is added to the environment from luminous volume particles
  - **Scattering**: when radiance heading in one direction is scattered to other directions due to collisions with the volume's particles

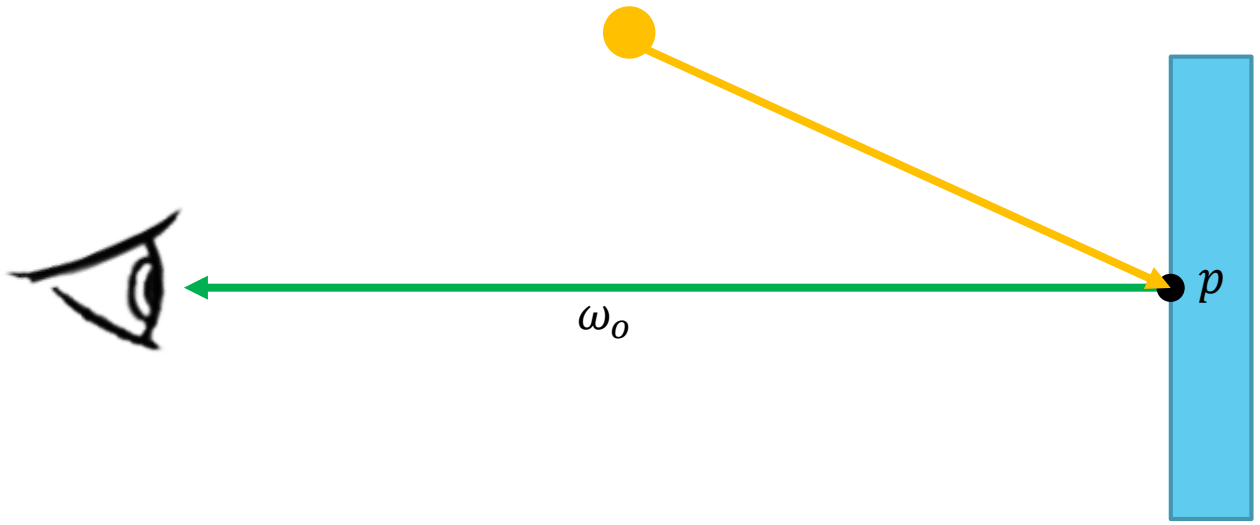
Seeing Through (one medium) – Examples



Simulating Volumes

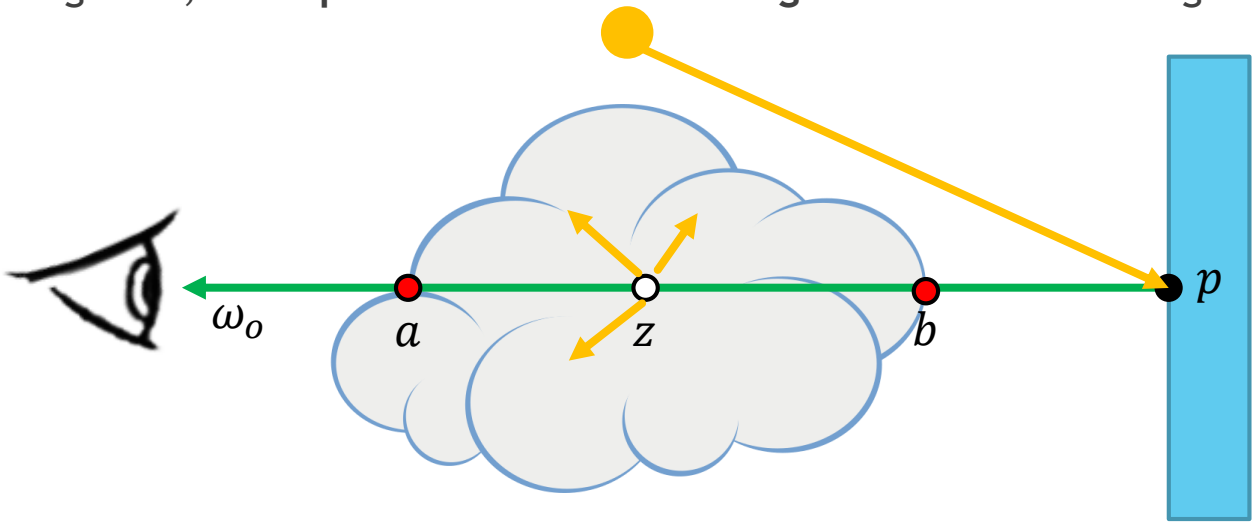
So Far: Assume Vacuum

- Approximate $L_o(p, \omega_o)$ through the rendering equation



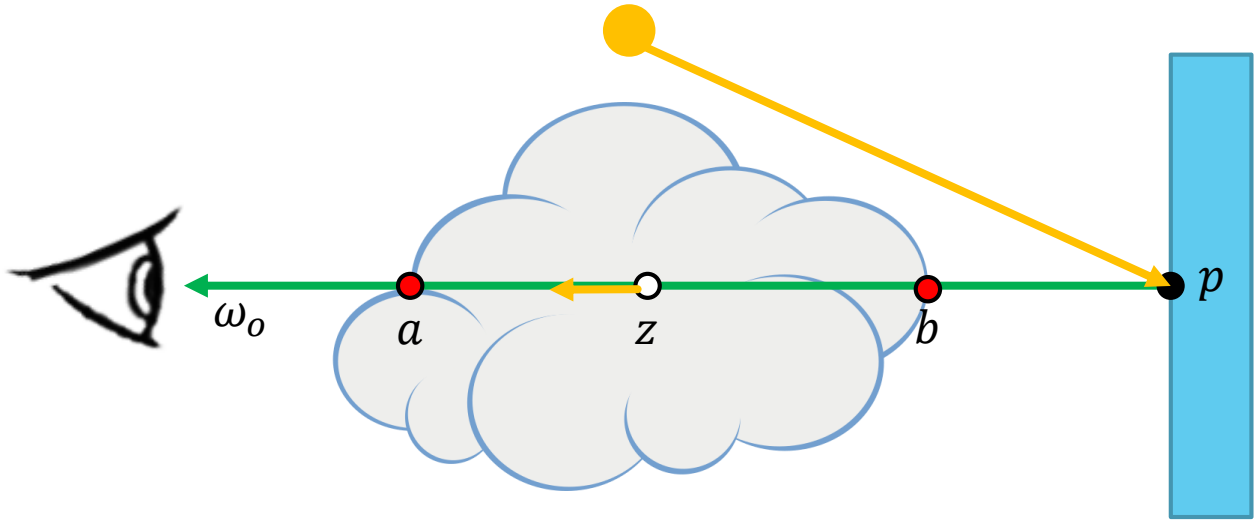
Participating Media: Attenuation

- Only a fraction $T(a, b)$ of the outgoing radiance $L_o(p, \omega_o)$ arrives at the eye
  - Any point z between a and b can absorb light
  - Any point z between a and b can out-scatter light
  - Together, **absorption** and **out-scattering** contribute to the light **attenuation** due to light-medium interaction



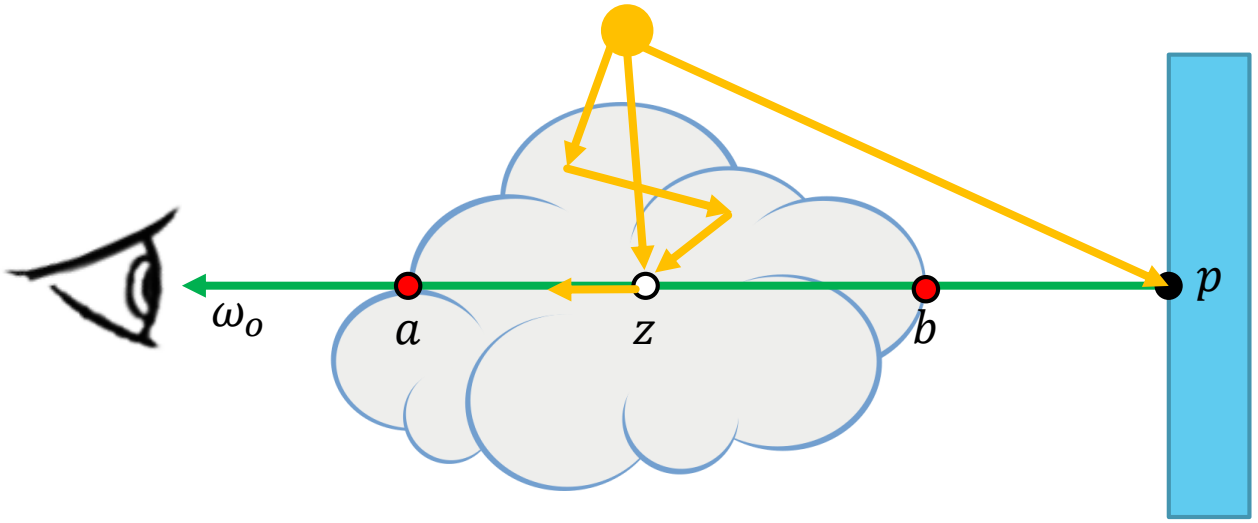
Participating Media: Emission

- However:
  - Any point z between a and b can also **emit** light



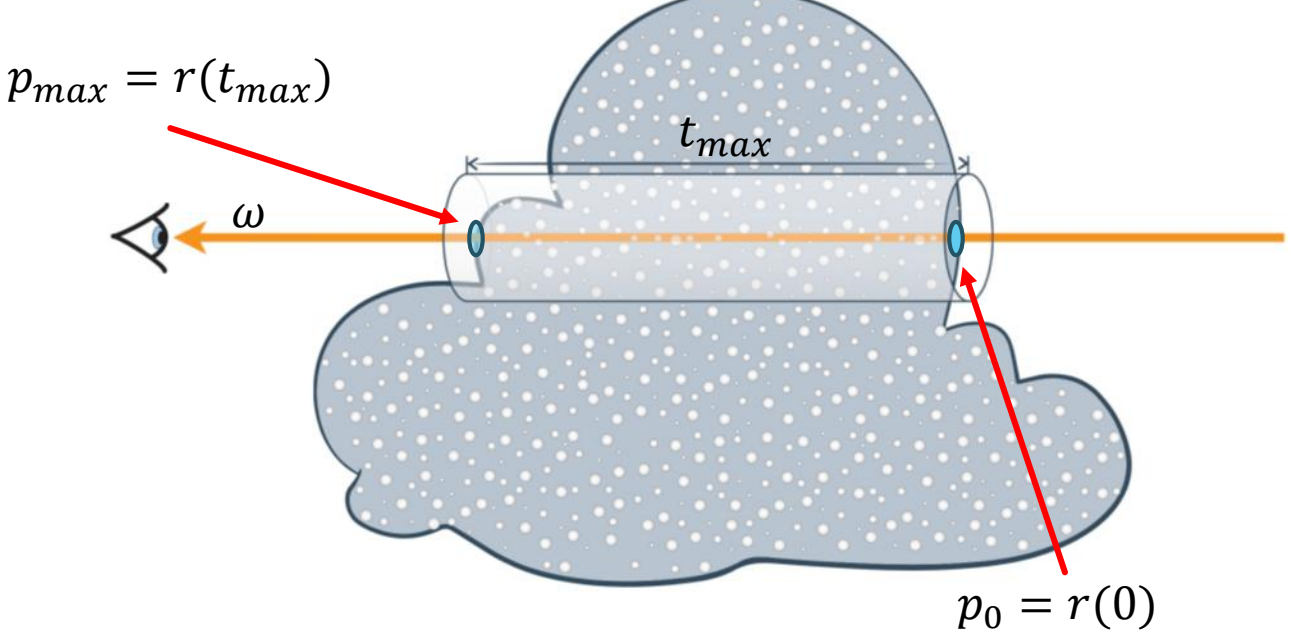
Participating Media: In-Scattering

- Moreover:
  - Any point z between a and b can emit light
  - At any point z, light might be incorporated onto the light path between p and the eye due to **in-scattering**



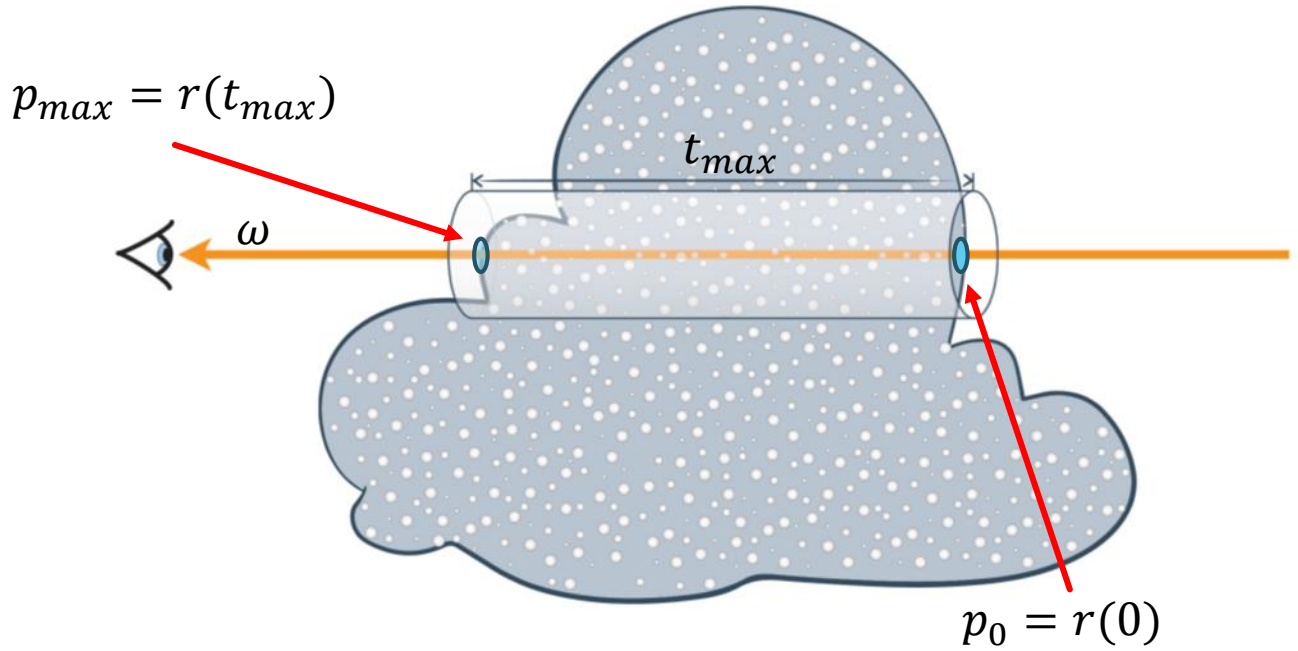
Problem Formalization

The Model: Set-up



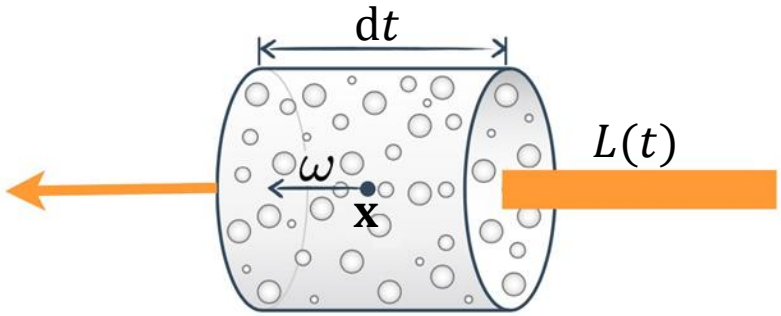
- A ray goes through the participating medium
- The medium is composed of particles which can *absorb*, *scatter* of *emit* radiance
- We are interested in the events occurring in the ray segment defined by
  - $r(t) = p_0 + t\,\omega$, with $0 \leq t \leq t_{max}$
  - t_{max} is the distance traveled by the ray inside the volume
- We assume that we know the amount of radiance B arriving at p_0 such that $L(0) = B$
  - Important for later: $L(0) = B$ is the boundary condition of our problem

The Model – Macro Level View



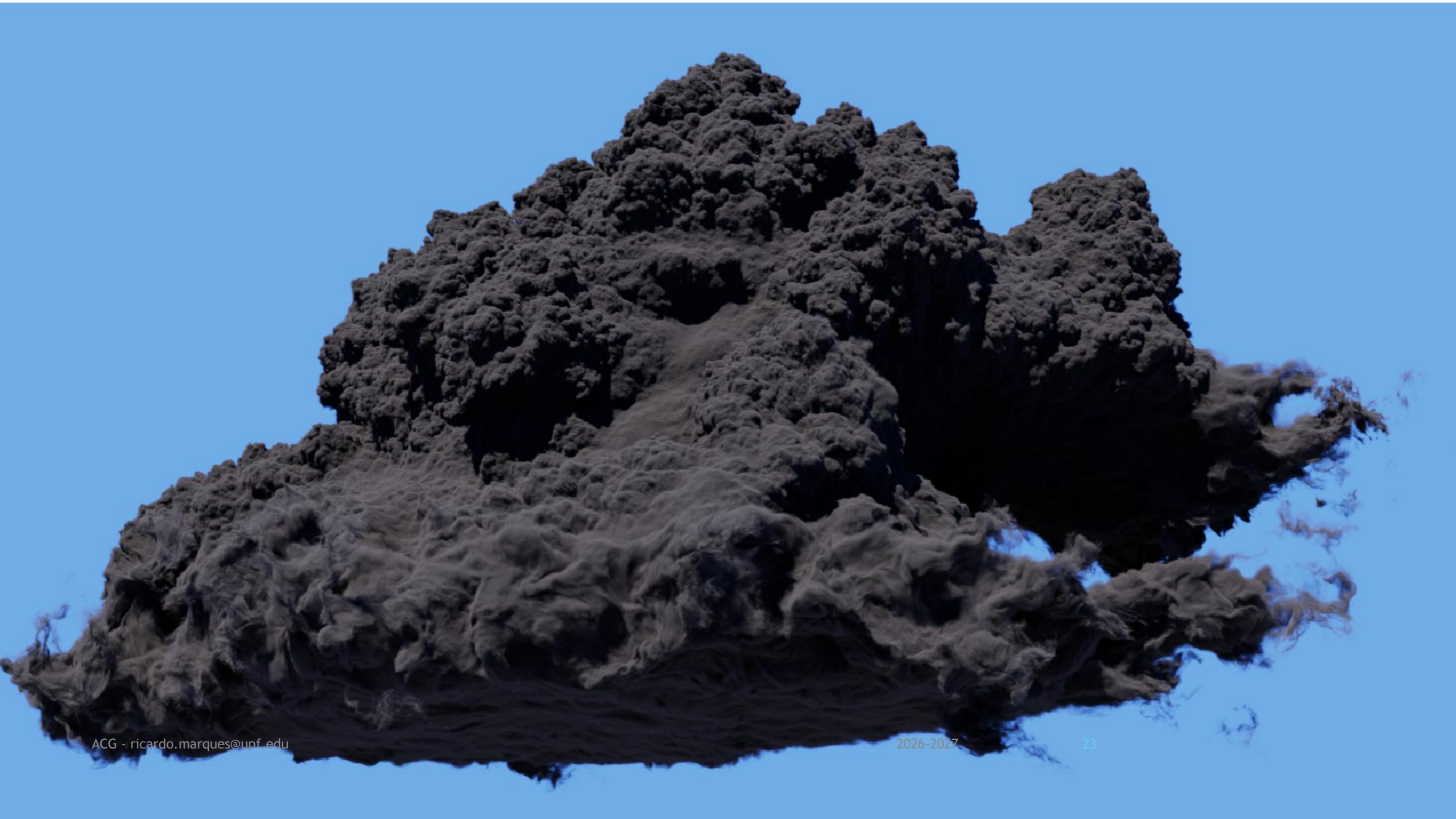
- Goal:
  - Express the amount of radiance which arrives to the eye, after going through the medium
  - Requires formalizing the notions of absorption, scattering and emission **(Micro Level)**

Absorption (Characterization)

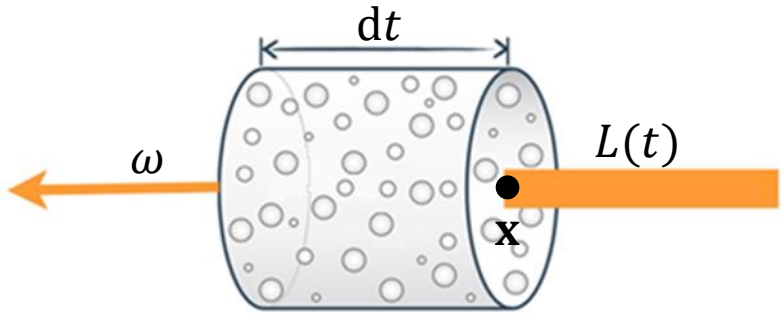


- Reduction in radiance due to the conversion of light to another form of energy, such as heat
- Characterized by the medium's **absorption coefficient** μ_a
  - Denotes the **probability density** that light is **absorbed** per **unit distance** traveled in the medium (m^{-1})
  - Sometimes specified as a function of both position and direction
    - $\mu_a(\mathbf{x}, \omega)$
    - For simplicity, we'll use $\mu_a(t)$
    - $1/\mu_a$ is the *mean free path*: the average distance at which an absorption event occurs
- Larger values of μ_a → more absorption → darker appearance
  - Intuition (linked to the notion of media density): denser → more probability of hitting an absorbing





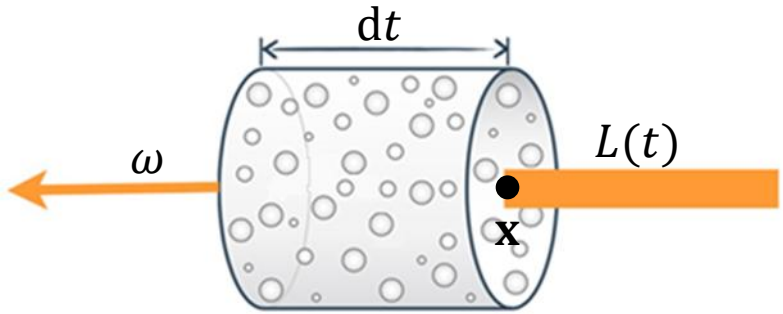
Absorption
(Formalization)



- Consider the above figure, where we have:
  - A differential cylinder filled with absorbing particles
  - A differential ray segment $\mathrm{d}t$
  - A point $\mathbf{x}$ along the ray such that $\mathbf{x} = r(t)$
  - A certain amount of radiance $L(t) = L(\mathbf{x}, -\omega)$ arriving at point $\mathbf{x}$ from direction $-\omega$

- **Question**: what is the change in radiance due to absorption in the differential volume?

Absorption
(Formalization)



- Consider the small ray segment dt
  - Along that segment, radiance is reduced (due to absorptions) by an amount ΔL, from L to L', such that:

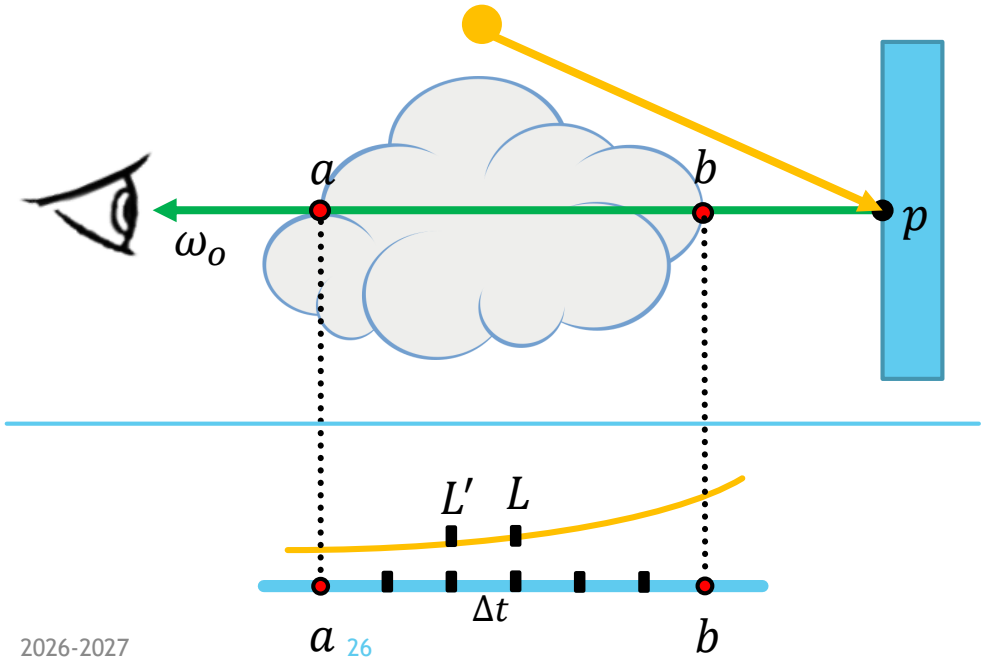
$$L' = L - (\mu_a \, \Delta t) \, L$$

  - Using $\Delta L = L' - L$, we can re-write this equality as:

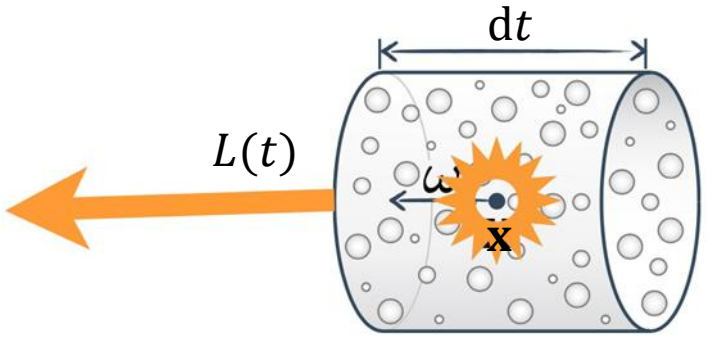
$$\Delta L = -\mu_a \Delta t \, \mathrm{L} \;\leftrightarrow\; \frac{\Delta L}{\Delta t} = -\mu_a \, \mathrm{L}$$

- For an infinitely small $\Delta t = \mathrm{d}t$, the above equation becomes:

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) = -\mu_a(t)L(t)$$



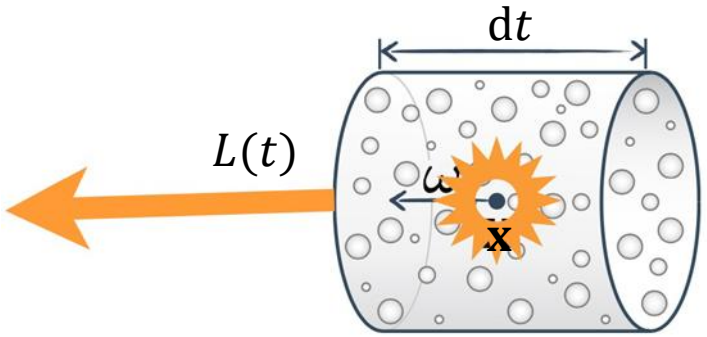
Emission
(Characterization)



- Increased radiance along a ray due to chemical, thermal, or nuclear processes that convert energy into visible light
- Characterized by the relation between how efficiently an object absorbs light (μ_a) and how efficiently it **emits light (L_e)**
- Emission is maximized when L_e is large while absorption μ_a is low



Emission (Formalization)

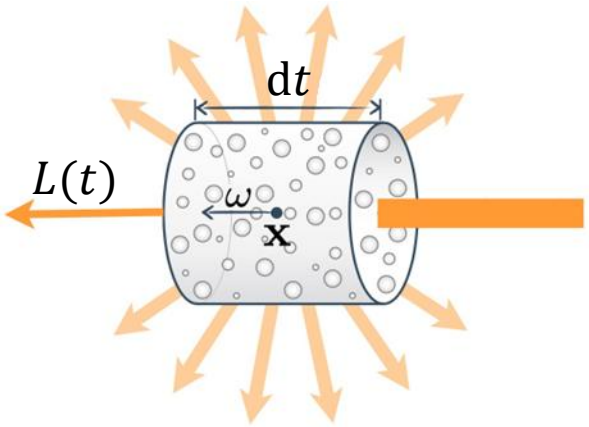


- Consider the above figure, where we have:
  - A differential cylinder filled with emissive particles
  - A differential ray segment $\mathrm{d}t$ with direction ω passing through the cylinder
  - A point $\mathbf{x}$ along the ray such that $\mathbf{x} = r(t)$
  - A certain amount of incident radiance $L(t)$ leaving point $\mathbf{x}$ towards direction ω

- Question: what is the change in radiance due to emission in the differential volume?

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) = \mu_a(t)L_e(t)$$

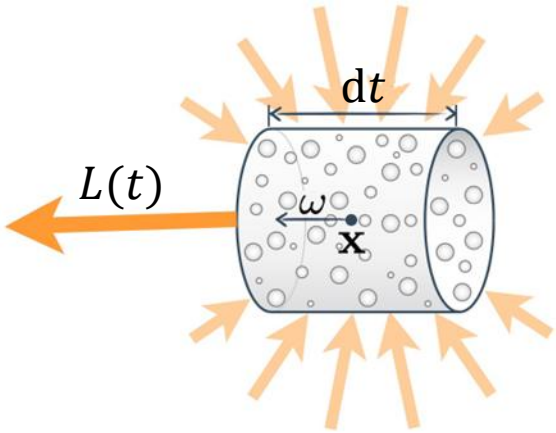
Out-Scattering



- Light that collides with particles and is scattered in different directions
- Leads to a reduction in the radiance exiting a differential region of the beam because some of it is deflected to different directions
- Also characterized by the scattering coefficient μ_s
- Change in radiance due to out-scattering in the differential volume:

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) = -\mu_s(t)L(t)$$

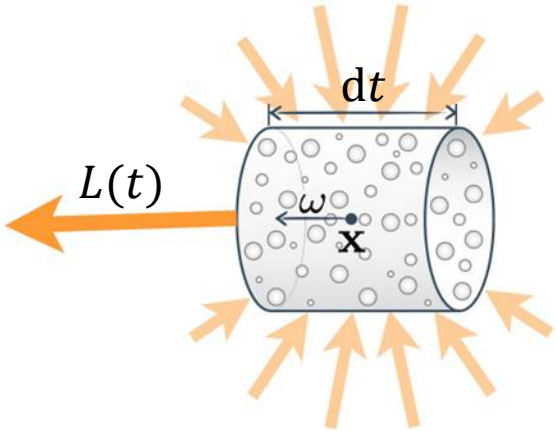
In-Scattering



- Radiance from other rays which is scattered into the path of the current ray
- Leads to an increase in the radiance exiting a differential region of the beam because some of it is incorporated from other directions
- Characterized by the scattering coefficient μ_s
- Change in radiance due to in-scattering in the differential volume:

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) = \mu_s(t)L_s(t)$$

In-Scattering



- Change in radiance due to in-scattering in the differential volume:

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) = \mu_s(t) L_s(t)$$

- L_s is commonly referred to as in-scattered radiance, and collects the incident radiance from all directions on the unit sphere, such that:

$$L_s(t) = \int_{S^2} f_t(\omega, \omega_i)\, L_i(t, \omega_i)\, \mathrm{d}\omega_i$$

where f_t is the **phase function** which characterizes the directional density of light scattered at a point t in a medium (i.e., the volume equivalent of the BRDF)





Summary of Light-Volume Interactions

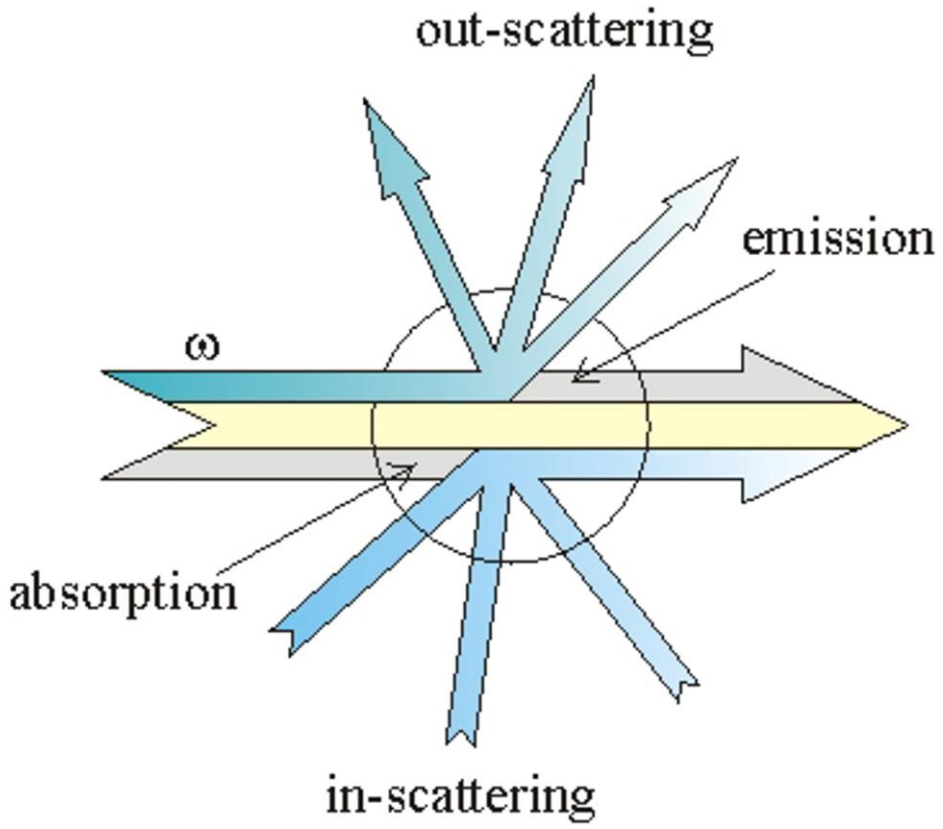


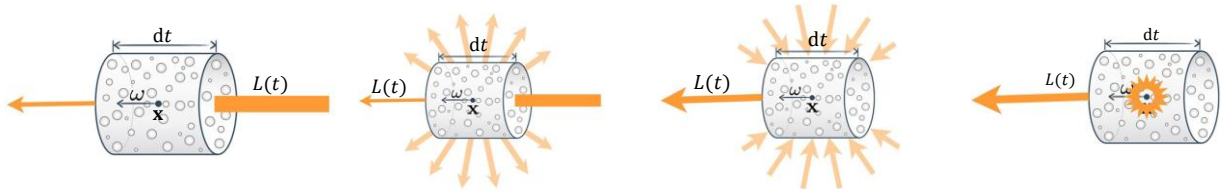
Image courtesy of Hendrik Lensch (MPI, WS07/08)

Radiative Transfer Equation (RTE)

Radiative Transfer Equation (Micro Level)

- The change in radiance traveling along direction ω through a differential volume element at particular point $\mathbf{x} = \mathrm{r}(\mathrm{t})$ is described by the radiative transfer equation (RTE, Chandrasekhar 1960)

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) = -\mu_a(t)L(t) - \mu_s(t)L(t) + \mu_s(t)L_s(t) + \mu_a(t)L_e(t)$$



Radiative Transfer Equation (Micro Level)

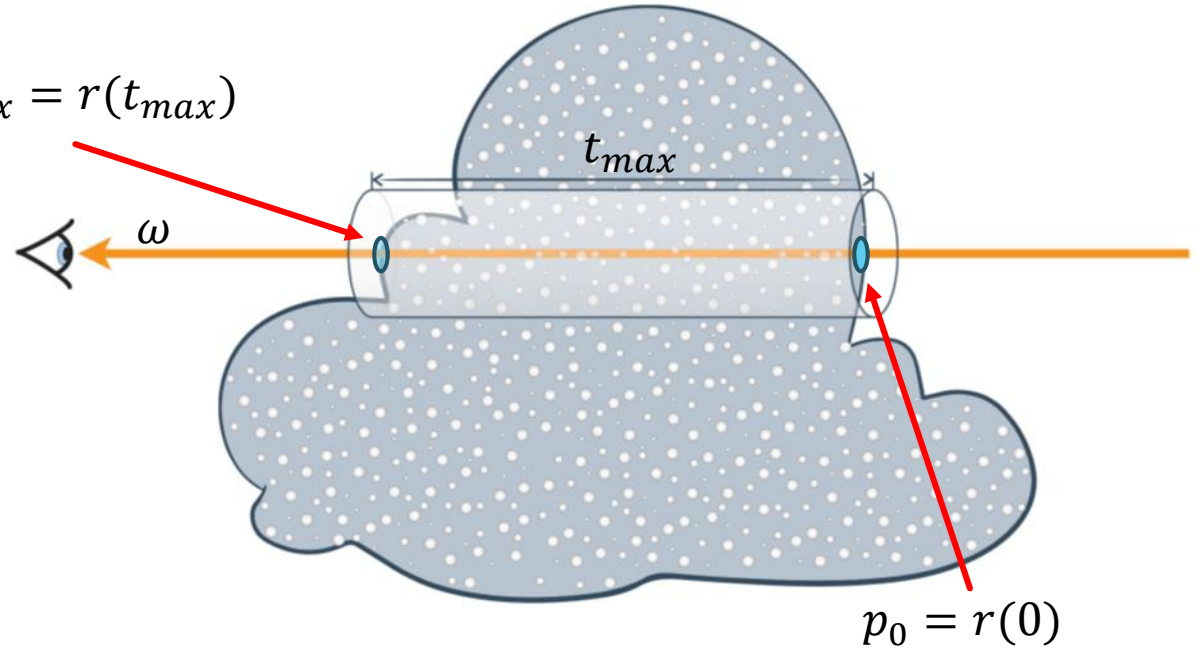
- Because we are at micro level, we only care about the change of radiance along the current ray
  - Direction of scattering is not relevant
  - Thus, absorption (μ_a) and scattering (μ_s) can both be described by a total extinction coefficient

$$\mu_t = \mu_a + \mu_s$$

- Using the above identity, we can re-write the RTE as:

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) = -\mu_t(t)L(t) + \mu_s(t)L_s(t) + \mu_a(t)L_e(t)$$

The Macro Level



$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) = -\mu_t(t)L(t) + \mu_s(t)L_s(t) + \mu_a(t)L_e(t)$$

- Gives us the change in radiance at each point in the ray
- We want radiance $L(t_{max})$ after interacting with all the volume (integral along the path)
- *Volume Rendering Equation*

In a Nutshell

- The micro-level light interaction with participating media is described by:
  - Absorption, when light hits particles of the volume which absorb light (reduction of radiance)
  - Emission, when the volume itself emits light (increase of radiance)
  - Out scattering, when light is scattered out of the ray path (reduction of radiance)
  - In scattering, when light is scattered towards the current ray path (increase of radiance)
- The in scattering event depends on the phase function, which characterizes how the volume “reflects” light (volumetric analogue of the BRDF)
- Using these concepts, we can set-up a micro-level model which
  - Radiative Transfer Equation, which describes how radiance changes in a volume due to the above-mentioned effects, considering the current volume features

From the RTE to the Volume Rendering Equation

The Absorption-Only Case

RTE and ODEs

- The Radiative Transfer Equation is a 1st order Ordinary Differential Equation

$$y'(x) + p(x)\, y(x) = q(x)$$

- Both the radiance function and its first derivative appear in the equation

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) + \mu_t(t)\, L(t) = \mu_s(t) L_s(t) + \mu_a(t) L_e(t)$$

- Our goal: solve the ODE to get an expression for $L(t)$ which does not involve $\frac{\mathrm{d}L}{\mathrm{d}t}(t)$

Simpler Models

- Rendering with participating media is a complex task
- Therefore, it is common that only a part of the effects are simulated
  - Leads to simpler and faster solutions
  - Leads to less realism
- In the following, we'll analyze a simple model
  - Only consider absorption
  - Use this assumption to derive a simple version of the Volume Rendering Equation
  - Use Ray Marching to approximate the VRE with a Riemann Sum

Absorption-Only Model

Absorption-Only Model Set-Up

- Only absorption is considered
  - $\mu_s(t) = 0$ ← (no scattering)
  - $L_e(t) = 0$ ← (no emission)
  - $\mu_t = \mu_a + \mu_s = \mu_a$

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) + \mu_t(t)\, L(t) = \mu_s(t) L_s(t) + \mu_a(t) L_e(t)$$

- The Radiative Transfer Equation is simplified to:

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) + \mu_a(t) L(t) = 0$$

- 1st Order Homogeneous ODE

$$y'(x) + p(x)\, y(x) = 0$$

Recall: Some Useful Identities

- Chain Rule

$$\big(f\big(g(t)\big)\big)' = f'\big(g(t)\big)\, g'(t)$$

- Derivative of Natural Logarithm

$$(\ln t)' = \frac{1}{t}$$

- Using the chain rule and setting $f(t) = \ln(t)$ and $g(t) = L(t)$, we get:

$$\frac{\mathrm{d}}{\mathrm{d}t}(\ln L(t)) = \frac{1}{L(t)}\,\frac{\mathrm{d}\,L}{\mathrm{d}t}\,(t)$$

Solving the Absorption-Only Model

$$\frac{\mathrm{d}}{\mathrm{d}t}(\ln L(t)) = \frac{1}{L(t)}\frac{\mathrm{d}L}{\mathrm{d}t}(t)$$

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) + \mu_a(t)L(t) = 0$$

Understanding the Absorption Model

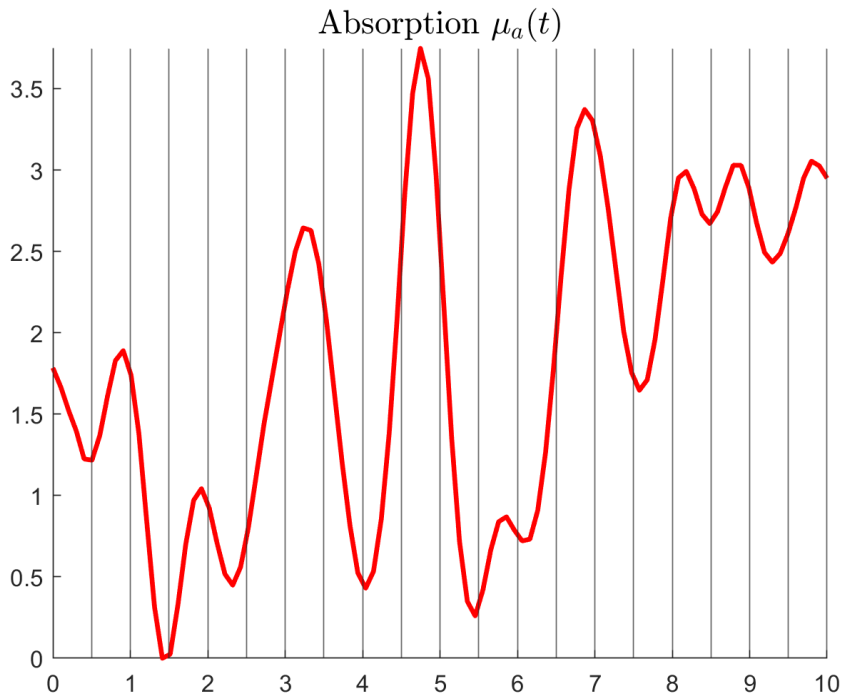
- The integral in the exponent is called ***optical thickness*** (τ)
  - $\tau(t_a, t_b) = \int_{t_a}^{t_b} \mu_a(t')\,\mathrm{d}t'$
  - Sums the attenuation μ_a coefficient from t_a to t_b
  - $\tau(t_a, t_b) \geq 0$
- The exponent term is called ***Transmittance*** (T)
  - $T(t_a, t_b) = e^{-\tau(t_a, t_b)}$ (Beer-Lambert Law)
  - $t_a \leq t_b$
  - Measures the fraction of radiance transmitted between t_a and t_b
  - $0 \leq T(t_a, t_b) \leq 1$

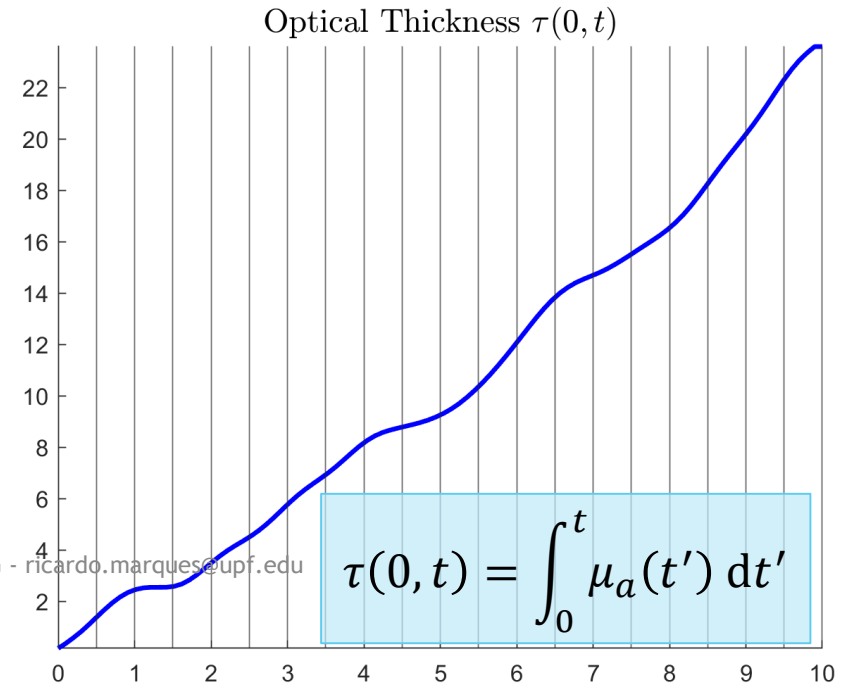
Different ways of writing the Absorption-Only VRE

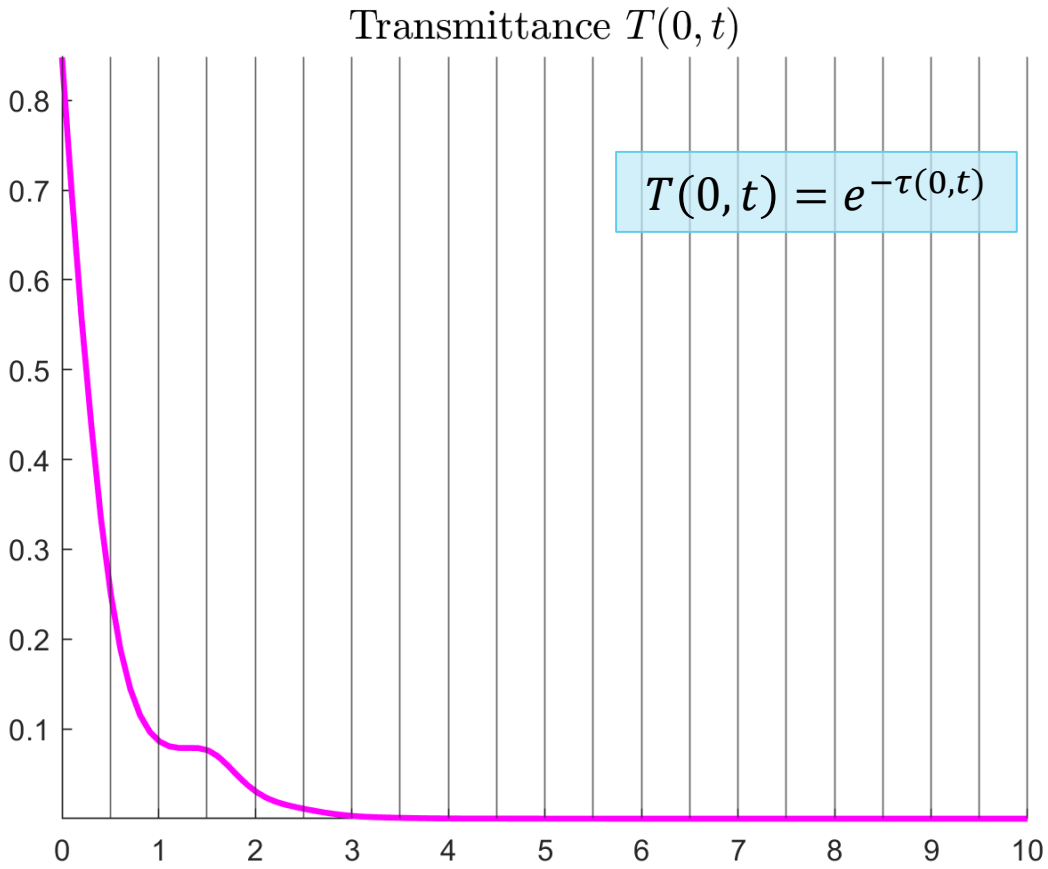
$$L(t) = B\, e^{-\int_0^t \mu_a(t')\,\mathrm{d}t'}$$

$$L(t) = B e^{-\tau(0,t)}$$

$$L(t) = B\, T(0,t)$$


Absorption $\mu_a(t)$
3.5
3
2.5
2
1.5
1
0.5
0
0 1 2 3 4 5 6 7 8 9 10


Optical Thickness $\tau(0,t)$
22
20
18
16
14
12
10
8
6
4
2
0 1 2 3 4 5 6 7 8 9 10
$\tau(0,t) = \int_0^t \mu_a(t')\,\mathrm{d}t'$

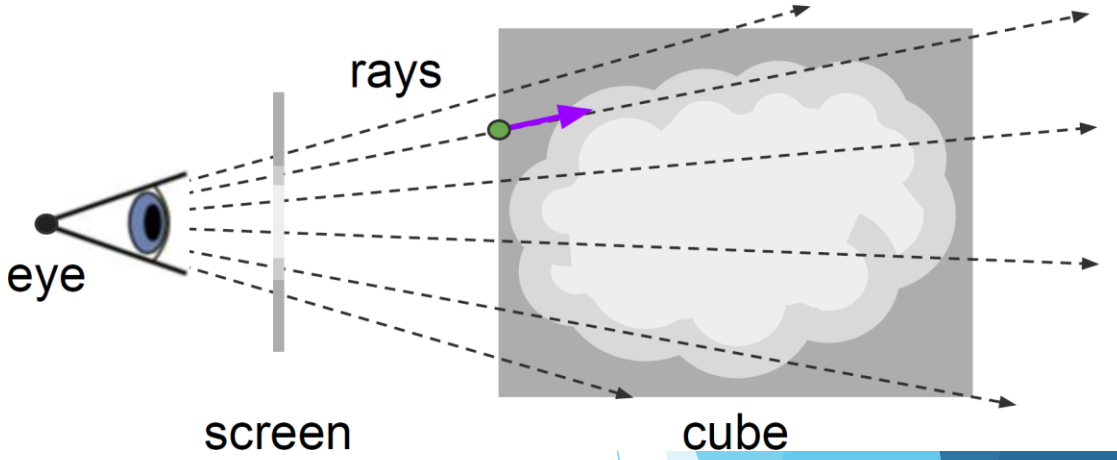

Transmittance $T(0,t)$
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0 1 2 3 4 5 6 7 8 9 10
$T(0,t) = e^{-\tau(0,t)}$

Approximation of the Absorption-Only VRE

A Ray Marching-based Approach

Approach

$$L(t) = B\, e^{-\int_0^t \mu_a(t')\,\mathrm{d}t'}$$



- To estimate the Absorption-Only VRE we will evaluate the optical thickness
- Two main families of estimation methods exist
  - Ray-Marching (see figure) and Quadrature Rules
  - Monte Carlo-Integration
- In this course, we will focus on *Ray Marching* (with a *Riemann Sum* as quadrature rule)
  - Conceptually simpler approach compared to the other methods
  - Well suited for beginners
  - Easier to implement

Homogeneous vs Heterogeneous Media

- A medium (or volume) is said to be **homogeneous** when its coefficients are constant
  - $\mu_a(t) = \bar{\mu}_a$, $\mu_s(t) = \bar{\mu}_s \rightarrow \mu(t) = \bar{\mu}_t$
  - The self emitted light is also constant ($L_e(t) = \bar{L}_e$)
  - Certain computations become simpler (more details in the following)

- In **heterogeneous** media, these quantities can vary
  - More memory consumption, but richer visual result

Optical Thickness with Homogeneous Media

- When $\mu_a(t) = \bar{\mu}_a$ is constant, then the optical thickness is:

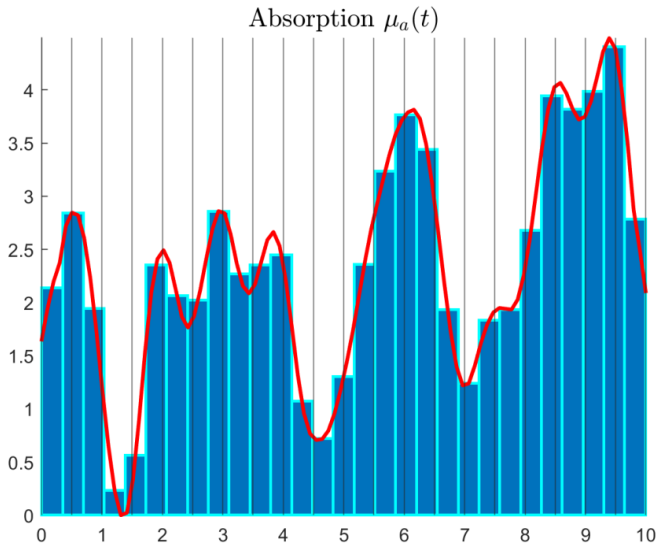
$$\tau(t_a, t_b) = \int_{t_a}^{t_b} \mu_a(t')\, \mathrm{d}t' = \bar{\mu}_a \int_{t_a}^{t_b} \mathrm{d}t' = (t_b - t_a)\, \bar{\mu}_a$$

- In our case $t_a = 0$ and $t_b = t$, and thus $\tau(0, t) = t\, \bar{\mu}_a$

- The optical thickness can thus be analytically computed (as well as the Transmittance):

$$L(t) = B\, e^{-\tau(0,t)} = B\, e^{-t\, \bar{\mu}_a}$$

Optical Thickness with Heterogeneous Media



- Assumption: we can *sample* $\mu_a(t)$ for any $0 \leq t \leq t_{max}$

- Then, the optical thickness integral can be approximated by a ***Riemann sum***:

$$\tau(t_a, t_b) \approx \sum_{i=0}^{N-1} \mu_a(t_i)\, \Delta t$$

where:

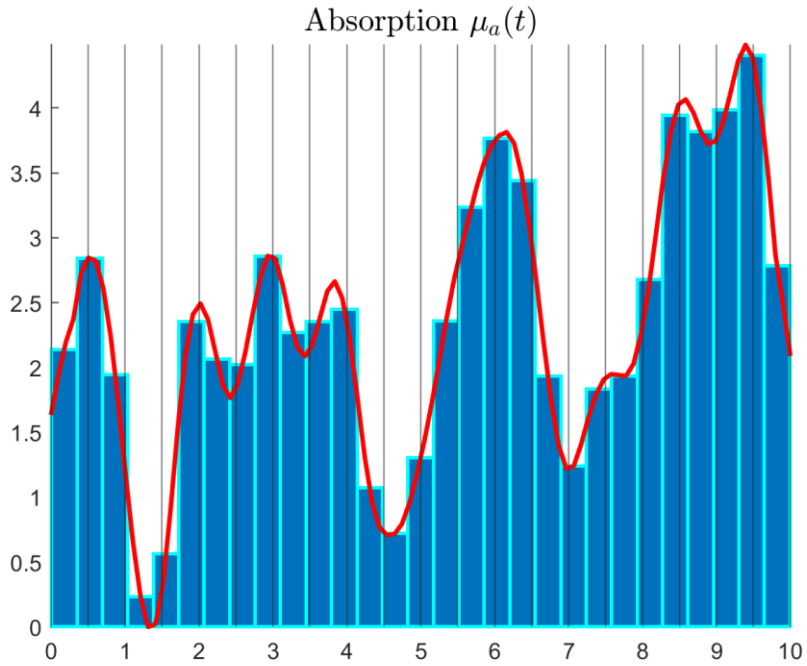
$\Delta t = \frac{t_b - t_a}{N}$ is the step size of the samples along the ray

$t_i = t_a + \left(i + \frac{1}{2}\right) \Delta t$ with $i \in \{0, \ldots, N-1\}$ are the N positions where $\mu_a(\cdot)$ is sampled

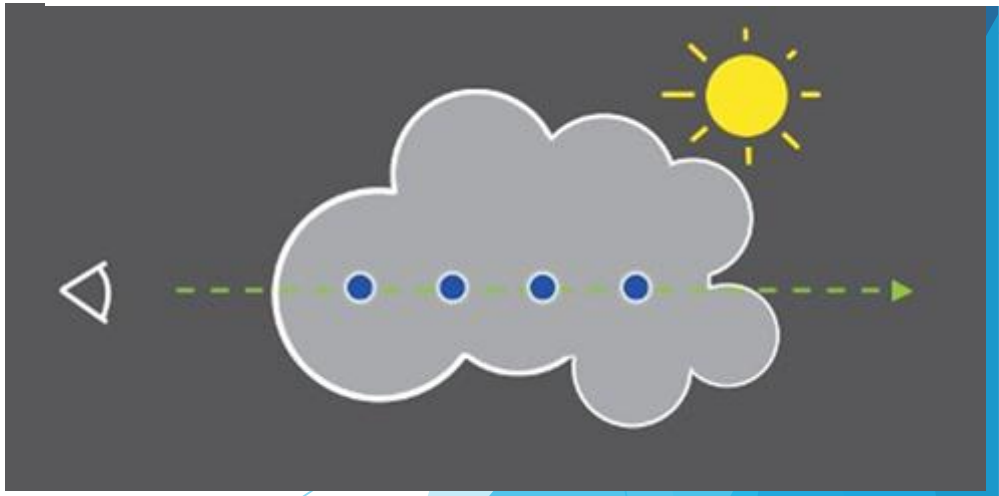
Optical Thickness with Heterogeneous Media

- Approximating optical thickness with a ***Riemann sum***:

$$\tau(t_a, t_b) \approx \sum_{i=0}^{N-1} \mu_a(t_i)\, \Delta t$$

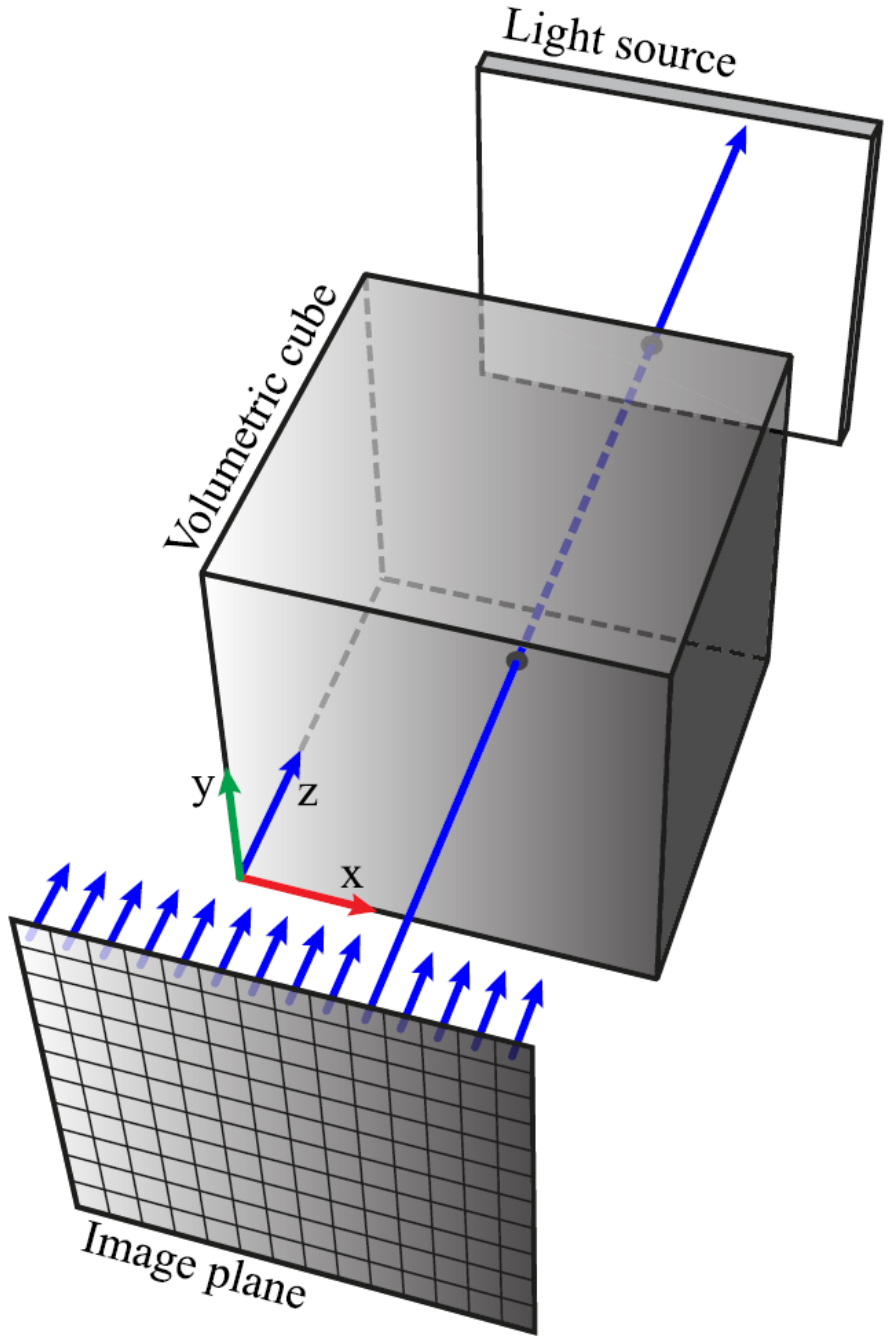


- Typically computed using ***Ray Marching***:
  - March along the ray
  - At each position t_i, sample $\mu_a(t_i)$ and multiply by the step Δt to get the rectangle's area

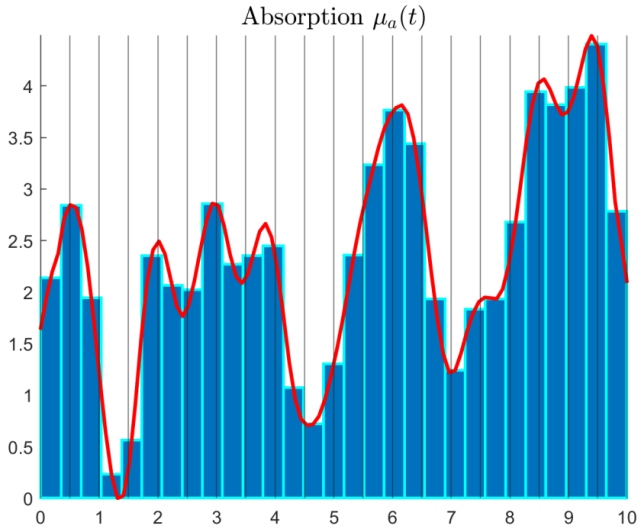


Ray Marching - Assumptions

- We'll assume that our scene contains a volume representing the features of a propagating medium (e.g., absorption and scattering coefficients)
- This medium is confined within the limits of a bounding volume
  - If its shape is a cube, we call it *Volumetric Cube*
- We can compute the intersection of a ray with the bounding volume
  - Yields entering and exiting positions t_0 and $t_{\max}$

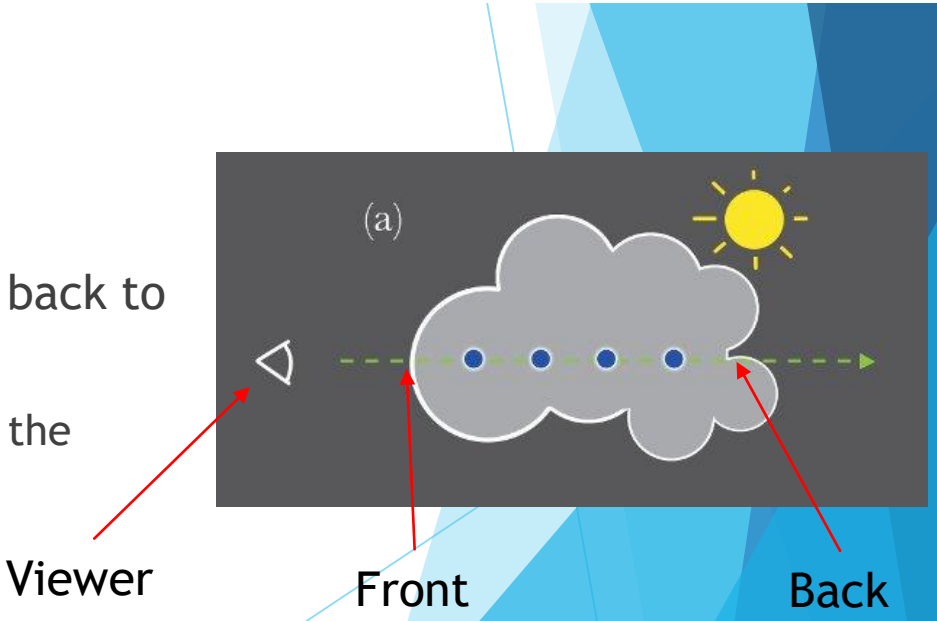


Back-to-Front and Front-to-Back



- As previously mentioned, we'll use Ray Marching to compute a Riemann Summation for approximating the optical thickness integral

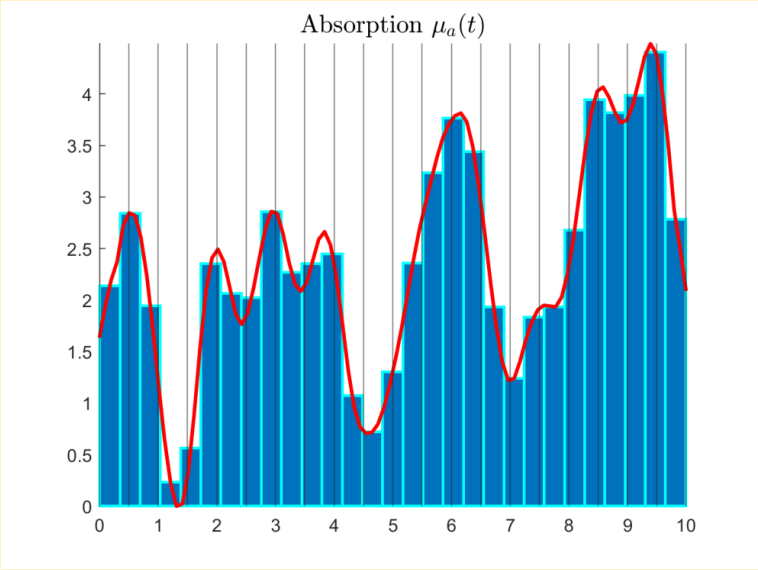
- Because summation is commutative, we can compute this sum back to front or front to back
  - Note: by convention, the back is the side of the volume that is on the opposite side of the viewer



Transfer Function

- In practice, the optical properties of the volume are not explicitly stored in the volume
- Typically, the 3D volumetric data stored is density, which needs to be converted to the optical properties of the volume (absorption, emission, …)
- This mapping between density at a given 3D location of the volume and optical properties is made by a *Transfer Function*

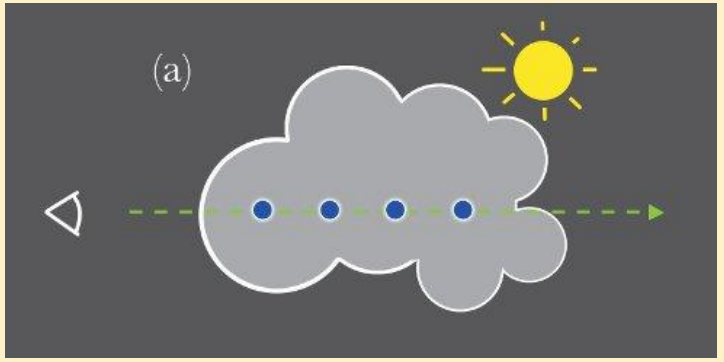
Ray Marching Loop for Absorption Model
Putting it all Together



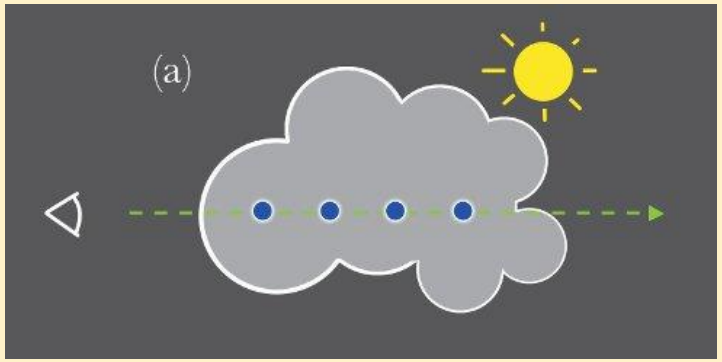
Ray Set-up

Traversal Loop

$$\tau(t_a, t_b) = \int_{t_a}^{t_b} \mu_a(t')\, \mathrm{d}t'$$



Brainstorming

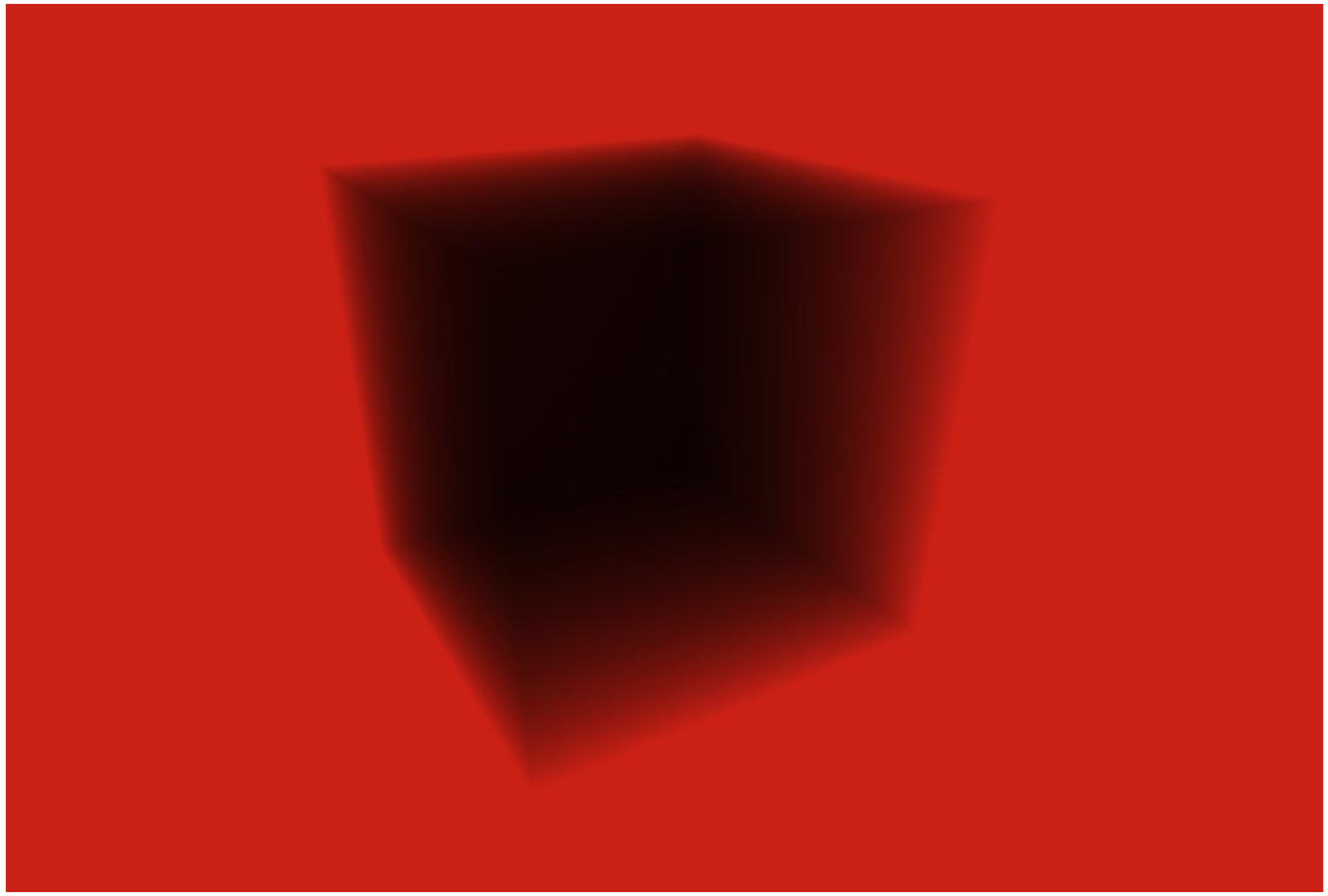


$$L(t) = B\, e^{-\tau(0,t)}$$

$$\tau(t_a, t_b) = \int_{t_a}^{t_b} \mu_a(t')\, \mathrm{d}t'$$

Assume you want to use ray-marching to compute the optical thickness τ and thus evaluate the amount of background light which is transmitted along a ray towards the viewer.

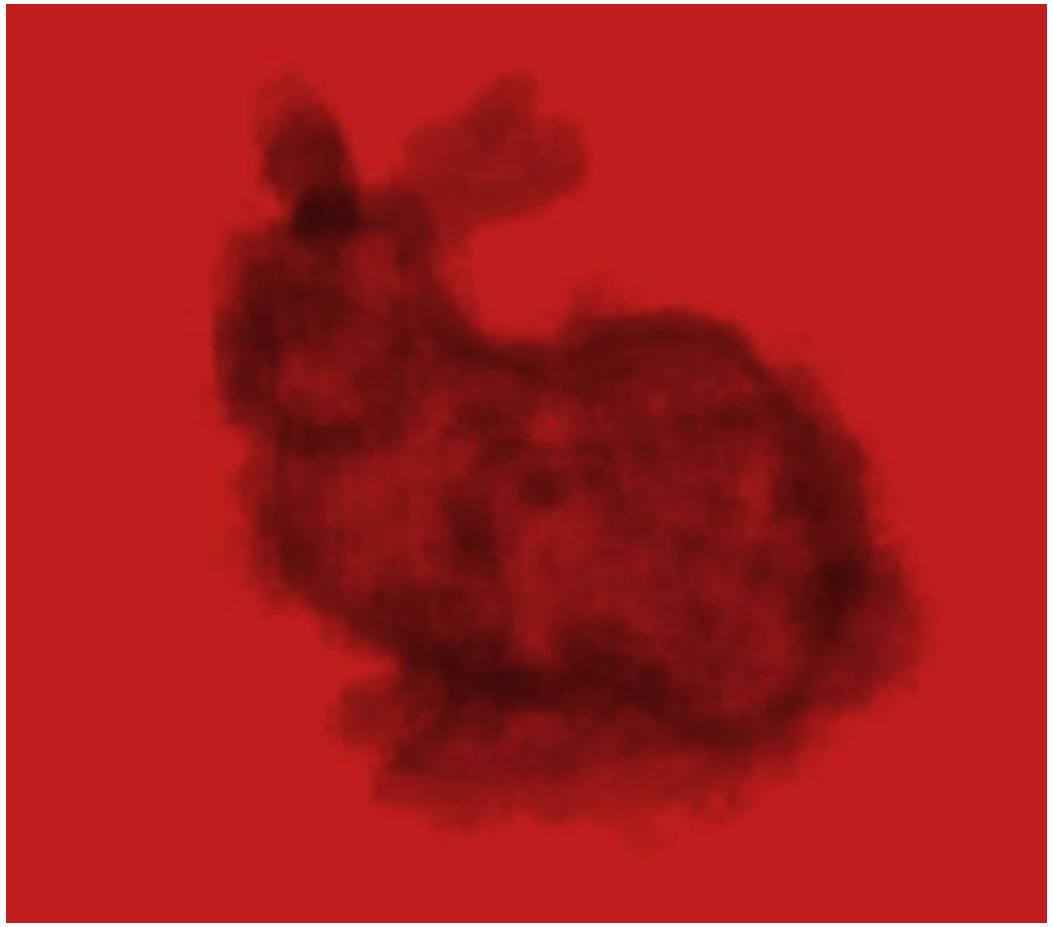
Which approach would you choose? Front to back or back to front? How do they compare?



Absorption Model Example Result Homogeneous Volume



Absorption Model Example Result Heterogeneous Volume



Absorption Model Example Result Heterogeneous Volume

Conclusion

- We have formulated and identified the RTE as a 1st Order ODE
- We formulated a simpler version of the Volume Rendering Equation
  - Absorption-Only
- We tackled the problem of providing an estimate to the Absorption-only VRE
  - Ray Marching & Riemann Summation
  - Introduced the concept of Transfer Function
- Next Class: towards the full Volume Rendering Equation (VRE)

Calendar

Week no.	Date	Theory	Practice	Seminar	Deliveries
Week 1	28 Sep 02 Oct	**T1**: Introduction	**P1**: Introduction to A1.1	-	
Week 2	05 Oct 09 Oct	**T2**: Basic Ray Tracing		**S1**: Work on A1	
Week 3	12 Oct 16 Oct	**T3**: Ray Casting, Recursive Ray Tracing	**P2**: Introduction to A1.2		
Week 4	19 Oct 23 Oct	**T4**: Rendering Equation, Monte Carlo Integration and Direct Illumination **T5**: Indirect Illumination and Path Tracing		**S2**: Work on A1	**D.A1**
Week 5	26 Oct 30 Oct	**T6**: Volume Rendering: Absorption-Only Model	**P3**: Introduction to A2.1		
Week 6	02 Nov 06 Nov	**T7**: Volume Rendering: Emission, Absorption and Scattering Model **T8:** Neural Radiance Fields and Topics' Presentation		**S3**: Work on A2	
Week 7	09 Nov 13 Nov	**T9**: **Midterm**	**P4**: Introduction to A2.2		
Week 8	16 Nov 20 Nov		**P5**: Work on A2	**S4**: Work on A2	**D.A2**
Week 9	23 Nov 27 Nov	**T10**: **1st Presentation**	**P6**: Work on FP	-	
Week 10	30 Nov 04 Dec	**T11**: **2nd Presentation** **(First Session)** **T12**: **2nd Presentation** **(Second Session)**		**S5**: Work on FP	**D.FP**

Credits

- Thank you to Josep Blat for making available the slides for ACG (2023)
- Thank you to Víctor Ubieto for the images with volumes with our framework
- Several slides and images are from the PBRT Book
- Thank you to Pascal Grittmann and Jaroslav Krivanek for making available their slides on Volume Rendering for PBR



The End

Questions?